BLUE ENERGY GENERATOR

Abdelrahman Ashraf Abdelrahman Mohamed Omar Saleh Saif Wael

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ABSTRACT

Egypt faces many challenges that negatively affect it in many areas. we are looking

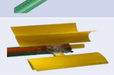
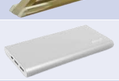
Moving forward to solve the energy problem. More than **1.5 billion people** in the world do not have access to electricity. This means that one in five people is forced to live without electricity. In our beloved Egypt, many solutions have been tried to ensure a better future for it Alternative energy. By searching for many solutions and found that there are many ways to generate energy, but the chose was through water. and that is done by We brought a **19-liter** water bottle and placed it on a stand made of wood. The bottle had a tube attached to it, there was a tap in the tube, and there was a turbine connected to the **dynamo** through a tube. In this project we generate energy through: First, we fill the bottle Secondly, we open the tap to direct the water to the turbine Third, the turbine rotates quickly due to the pressure of water falling from the tap Fourth: Because the **dynamo** is connected to the turbine, this leads to power generation through the **dynamo** We identified the design requirements were high efficiency, and environmentally friendly. After by implementing the test plan, it was found that our prototype achieved good results, The project produces **6.14 volts in 5 min** and multiply it in **0.085 ampere** to produce **156 joule**. , which coincidentally meets the design Requirements are fully met. Our project was built step by step, with great care.

INTRODUCTION

Essential challenges are faced by Egypt, including those of alternative energies, overpopulation, scientific and technological advancement for all, and climate change. Energy is considered crucial for the future of humanity, with electricity being the most widely used form. As a result, recent years have seen a growing interest in generating electricity from alternative sources **(1)**. Overpopulation is caused by an abundance of services in a specific area, which negatively affects another population where such services are not available. This, in turn, necessitates the even distribution of energy sources throughout the country **(3)**. Hydropower technology is used to generate electricity. Hydropower is a beneficial method due to its ability to reduce harmful gas emissions, which are a primary cause of global warming **(2)**. The Guri Dam in Venezuela is a **hydroelectric power** plant. It boasts advantages such as electricity generation through **20 generating** units capable of producing up to **10,235** megawatts. A complex computer-controlled system ensures the safe and efficient operation of its turbines and gates. While providing **75%** of Venezuela's electricity, it does have drawbacks such as environmental disturbances and sedimentation caused by reservoir management practices. The dam's activity has negative impacts on the environment and surrounding population. The Wadi Dayaqah Dam also contributes to generating alternative energy. The dam utilizes **hydroelectric power** for electricity generation, with both advantages and disadvantages. **Hydroelectric power** is a low-cost, renewable energy source that is sustainable and helps to reduce flood risks. However, challenges exist such as limited water access due to unsuitable dam locations **(5)**. The dam employs a water reservoir located above a school building for electricity production. This solution is designed to address specific requirements by generating electricity quickly, being environmentally friendly, and focusing exclusively on the school environment. The prototype is a small system modeled after the existing water system in the school. It is made up of two ordinary water bottles connected by special water pipes. Water flowing from the pipes falls towards the generator's turbine. Based on previous research, five blades are considered an optimal number for a water turbine **(4)**. Therefore, the **dynamo**'s turbine has five blades in order to rotate the magnet as efficiently as possible and produce power.

MATERIALS

Table (1)

Product	Source	Price	Image
19 L water bottle	From home	0 LE	
Wood	Carpenter	150 LE	
Tubes	Plumber	25 LE	
Blades	Electronics shop	50 LE	
Dc motor	Electronics shop	250 LE	
Power bank	From home	0 LE	
3V LED	From home	0.5 LE	

Total cost: 475.5 LE

METHODS

In any project in our life, some steps should be followed to ensure the construction of the prototype and achieve the required purpose of the project; For the prototype's body; A hole underneath the plastic bottle was made to make a space for the tube, after that some wax was used and burning glue to close the openings in the bottle, respectively we bought a wooden stand to hold the bottle and the tube together; For the turbine we brought an old plastic tube and a roller bearings for it; For the **dynamo** we brought an old DC motor and installed a neodymium magnet inside it to increase its efficiency as a **dynamo**.

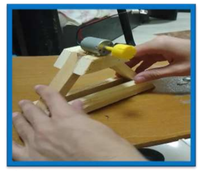


Fig (1)

Test plan:

In the first test plan, readings of volts and amps were taken, but the bottle leaked water because of the jaw glue. Despite all this, the reading was **1.5 volts** and **0.02 amps**.

The second test plan learned from our mistakes in the first test plan and corrected it, so this led to the success of the test plan, but it read **3.37 volts** and **0.05 amps,** which was small and does not follow the design requirements, and this was due to our use of spoons in the manufacture of turbines.

In the third test plan, blades were replaced instead of spoons for the turbine, and this led to the success of the test plan, and the readings were **6.14 volts** and **0.085 amps**, which follows the design requirements.

Fig (2)

Fig (3)

RESULTS

The Trail 1: The **dynamo** produces enough **1.5 Volt 0.02Amperes** to light a lamp. That was when the spoons were used to make a turbine and connected the dynamo with the turbine by a long pipe.

The Trail 2: Spoons were replaced with blades which we made the turbine and the blades. The ideal number of blades was **5 (4)**, and the **dynamo** produced more than a **3.37 Volt 0.05 Ampere**, which is enough to light a bulb.

The Trail 3: water and pressure were increased by opening the tap as wide as possible, and this led to an increase in the **Volts by 6.14 and the Ampere increase by 0.085**. This is the highest readings concluded for the volts and amps, as illustrated in fig (4) (5) and table (2).

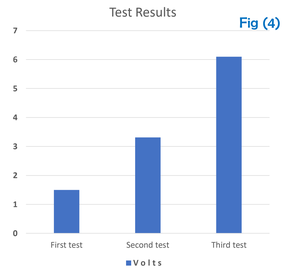


Fig (4)

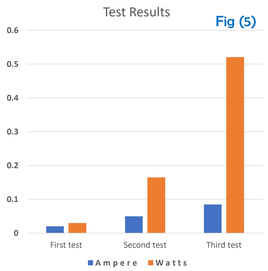


Fig (5)

Table (2)

Unit	Volts	Ampere	Watts	Joule
Frist test plan	1.5 Volts	0.02 Ampere	0.03 Watts	9 Joule
Second test plan	3.3 Volts	0.05 Ampere	0.165 Watts	49.5 Joule
Third test plan	6.14 Volts	0.085 Ampere	0.521 Watts	156.5 Joule

ANALYSIS

Egypt faces a demand increment for electricity, by dint of the overpopulation and overuse of fossil fuels, this increases the aquatic pollutants as it also causes climate change **(1)**. For the reason that usage of fossil fuels is directly proportional to the pollutants appearance. In such a way that overpopulation varies directly with alternative energy resources distribution, as residents concentrate in the places of energy resources concentration **(3)**.

Hydroelectric power presents a significant solution for the renewable energy problem, on account of reducing fossil fuels consuming, climate change and overpopulation, as the modern technology used in functionality of the generator is progressive **(5)**. This system demonstrated a significant increase in the energy generation compared to the traditional ways, as the DC motor was converted into a **dynamo** by reflecting the wires and installing neodymium magnet inside it, also the flow rate was calculated to be **0.12 L/s (PH 1.08)**; Although the project had positive outcomes, for instance the high quantity of generated electricity and is easily usable in generation, in addition to that, the project is non-implement; but, also negative outputs were encountered, for example, a leaking in the water tube caused losing of some stored water, also a curve in the blade which lead to stopping of the motion of the turbine; After the test plan data has been analyzed, the success of the project has been concluded as it met the design requirements of energy production, ease of usage, friendly of materials and storage of electricity; Additionally, the project met the required quantity of energy which was **150 Joule**, nevertheless the project produced **156.57 Joule**.

Based on the construction methods of our prototype, a test plan was implemented and diverse data was recorded to meet the design requirements, the initial investigation focus of the project was on the potential difference (V) and current intensity (I) outputs, as shown in fig (6) and fig (7) throughout the voltmeter and ammeter, which readings were **6.14 volt** and **0.085 ampere**, in addition to that the power (P) which is calculated in the relation **Power $= IV$** that equals **0.52 watt**, so that energy (E) can be calculated by employing the rule **$E = Pt$** which is equal to the rule **$E = IVt$ (ES 1.09)**, so that the energy will be **156.57 Joule**.

Fig (6)

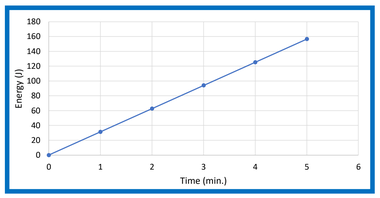


Fig (8)

Fig (7)

As illustrated in fig (8), the relation between the energy (E) in joule and time (t) in minutes, which is presented in a linear function.

It also represents the quantity of the energy output per minute, which equals 31.314 J/min, as the test plan continued for five minutes, so that the total energy output throughout the project equals $total\ energy = \frac{E}{t} \times total\ time$; Therefore, total energy equals **31.314 × 5 = 156.57 joule**.

Table (3)

Time (min.)	0	1	2	3	4	5
Energy (J)	0	31.314	62.628	93.942	125.256	156.57

As explained in **table (3),** the data outputs calculated by the rule $E = IVt$ were collected in a table between energy (E) in joule and time (t) in minutes.

This table indicates the change in the energy per minute (J/min), also it was increasing constantly with a rate of **31.314 J** each minute, to give a final energy of **156.57 J**.

After all the calculations and results collection, the project has achieved all the design requirements of the quantity of Joules, ease of usage, environmentally friendly materials, non-impediment, fast energy production, storage of electricity and beneficial usage of it.

Learning outcomes:

Table (4)

LO	Description
ES 1.09	Comprehending electricity and its generation ways through the generators.
MA 1.07	Comprehending linear and quadratic functions in order to presenting graphs.
CH 1.09	Comprehending and determining the physical properties of the materials.
English	For documenting all the information in the poster.
PH 1.08	Calculating the volume flow rate of the water.

CONCULSION

After the results and analysis written. This idea taken from the prior solution and changed in it The. Solution that chosen to the project is successful after three trials. Because achieved the required results. The required results are **150 joules** in maximum **5 minute** then , multiply the ampere in the volt that exerted to get the joule. Already the project of my team achieved this result. The project produces **6.14 volts in 5 min** and multiply it in **0.085 ampere** to produce **156 joule**. The prior solutions that found are that produce energy from the solar energy, wind, water. My team used the water to round the turbine and the **dynamo** that the team made produce energy. But by working on the idea of the water and evolution this idea by strong the push of water to produce more energy and use the material depend on design requirement that material is eco-friendly and low cost and not pollute the environment. Such, as using the wood that not pollute the environment. Lastly, from the important factor that evolution is the **dynamo** that made is very strong and he push of water stronger and used the able to move the **dynamo**. This is from the important factor that the team evolution it.

Recommendation

Perfecting a project requires attention to detail and improvement in various aspects. For future students, here are some key enhancements to consider:

Enhanced Magnet: Explore advancements in magnet technology to improve dynamo efficiency. Consider utilizing Neodymium Permanent Magnets, known for their strength and effectiveness in dynamo construction.

High-Efficiency Copper: Opt for high-quality copper, such as Oxygen-Free Copper, for the construction of electrical coils and wires in dynamos. This material enhances conductivity and overall performance.

Specialized Plastics: Utilize plastics optimized for water turbine manufacturing, such as Polyamide (Nylon). These materials offer durability and efficiency in turbine operation.

Optimized Tube Design: Incorporate a tapered tube design to increase water pressure, thereby enhancing turbine performance. Gradually reducing the tube's opening helps maximize the energy transfer to the turbine, improving overall efficiency.

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For Further Information

Omar Saleh: Omar.2223041@stemsharkya.moe.deu.eg

Abdelrahman Ashraf: Abdelrahman.2223029@stemsharkya.moe.deu.eg

Abdelrahman Mohamed: Abdelrahman.2223032@stemsharkya.moe.edu.eg

Saif Wael: Saif.2223026@stemsharkya.moe.deu.eg